Design Challenge 1

Digitally Controlled VGA

Part 1: OTA Design

Required Specifications:

Technology	65nm
Supply voltage	2V
Reference current seed	Use a single 20uA ideal reference current source.
Output type	Single ended
External load	500 fF
Feedback type	Capacitive (use 500fF unit capacitors ONLY)
Digital control	Single bit signal (D0)
Closed loop gain	D0 = 0: 6 dB D0 = 1: 12 dB
LG PM (worst case)	> 60o
DC LG	> 54 dB
Closed loop BW	> 10 MHz
Output swing	> 1.6 V pk-to-pk
Power Consumption	Minimize

Design Observations:

- Required Loop gain is large: 2 Stage OTA works better
- Digitally controlled: A switch is needed
- Single Ended
- Wide Output swing: I need Vout at middle of VDD (1V) and Vin at the same voltage using feedback thus nmos input stage is best to sustain the bigger input voltage.

A Miller 2 Stage OTA works best of this design with capacitive feedback and a switch to control the gain.

- No spec on CMIR: Use NMOS for the NMOS stage and PMOS for the second stage
- Use Inverting configuration for better performance

Design Parameters:

$$LG = 54dB \rightarrow 501.187 \approx 500 = \beta A_{OL}$$

$$Worst\ Case\ \beta = \frac{1}{5} \rightarrow A_{OL_{worst\ case}} = 2500 \rightarrow 68dB$$

$$BW_{OL} = \frac{BW_{CL}}{1 + LG} \approx 20KHz \rightarrow GBW_{OL} = 50MHz$$

Worst case Cout:

$$C_{OUT} = C_{in} + C_L = 2.5pF$$

$C_C = 0.5 * C_{OUT} = 1.25pF$

Input Transistors:

$$A_{OL} = 68 \rightarrow divide\ into\ 2\ for\ both\ stages$$
$$A_{OL}(First\ Stage) = 34dB \rightarrow 50$$
$$GBW_{OL} = \frac{gm_{input}}{2\pi C_C} \rightarrow gm_{input} \approx 400\mu S$$

$$A_v \approx gm_{input} \frac{ro}{2} = 50 \rightarrow ro \approx 255K\Omega$$

Assume MI $\frac{gm}{I_d} = 15$

$$\frac{gm}{I_D} = 15 \rightarrow I_D = 26\mu A$$

$$Assume\ I_D \approx \mathbf{40\mu A}\ higher\ margin\ for\ the\ big\ BW\ required\ and\ easier\ mirroring$$

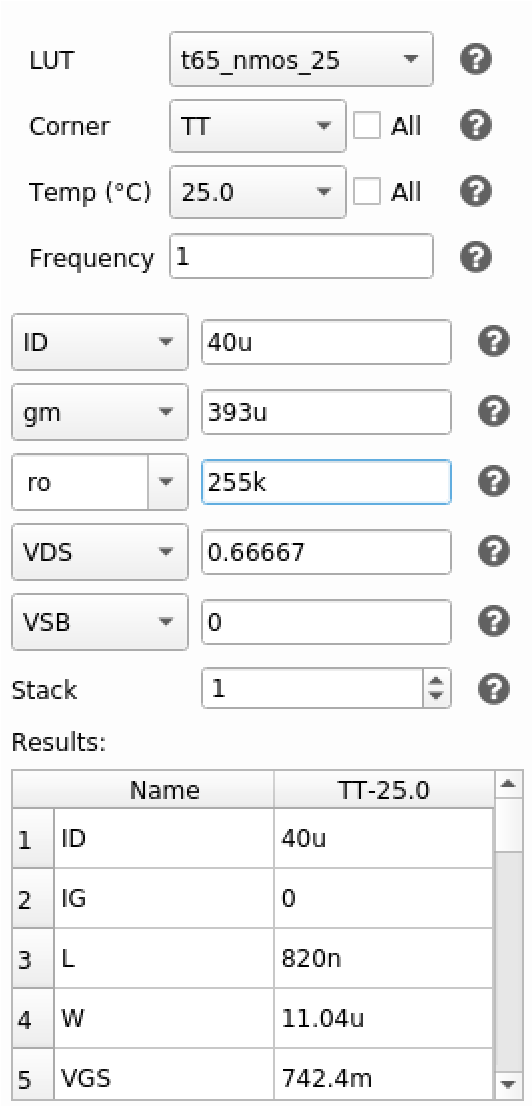

Figure 1 Input pair Sizing from ADT SA

$$\boldsymbol{L = 820nm \quad , \quad W = 11\mu m}$$

PMOS Current Mirror Load:

From previous device:

$$I_D = 40\mu A \quad , \qquad ro = 255K\Omega$$

Since there is not spec on the CM input range, I'll assume moderate inversion:

$$\frac{gm}{I_D} = 15$$

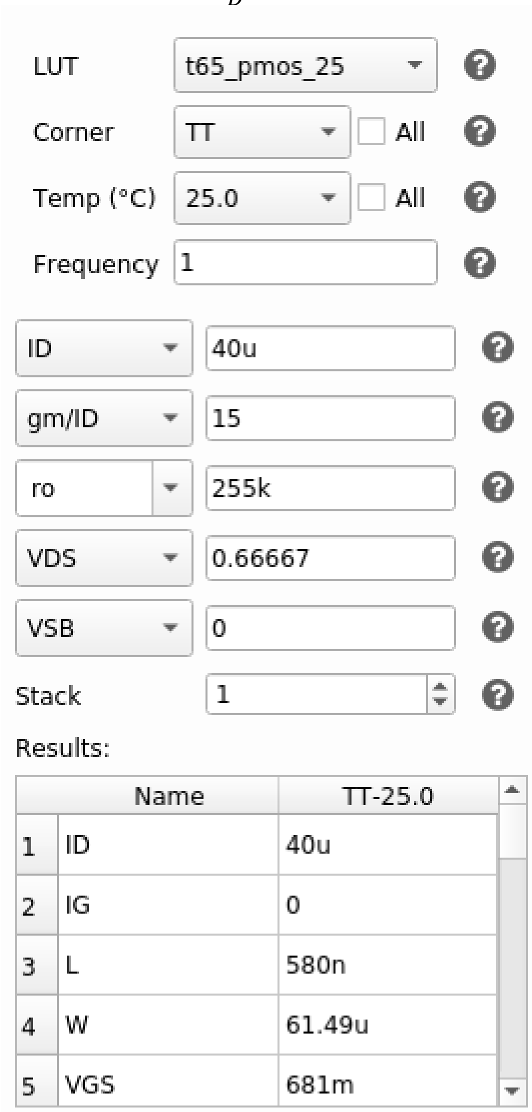

Figure 2 Sizing of PMOS Load from ADT

$$\boldsymbol{L = 580nm \quad , \quad W = 61.5\mu m}$$

Tail Current Mirror Sizing:

From Previous Parts:

$$I_D = 80\mu A$$

$$Assume\,\frac{gm}{I_D} = 10\ (Weak\ Inversion)$$

$$Assume\ Long\ L = 1\mu m\ (For\ accurate\ Mirroring)$$

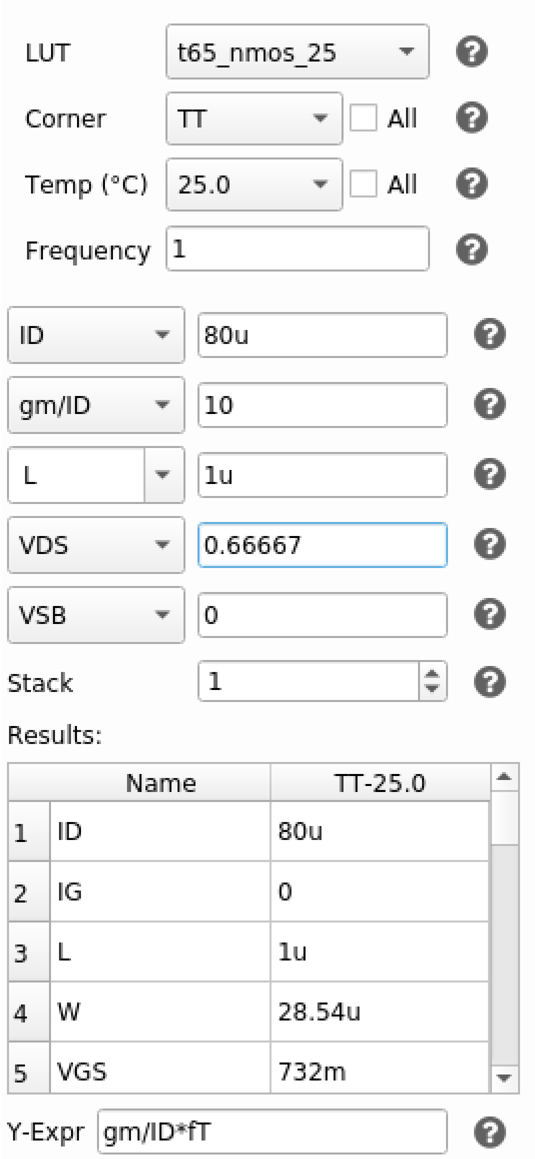

Figure 3 Sizing of Tail Current Mirror from ADT

$$\boldsymbol{L = 1\mu m\quad ,\ W = 28.54\mu m\,,ro = 143K\Omega}$$

Current Mirror:

For the Tail current Mirrors I just used multipliers to get accurate and quick mirroring of the currents in each stage.
I assumed $I_{B2} = 4 * I_{B1}$ for better Stability.

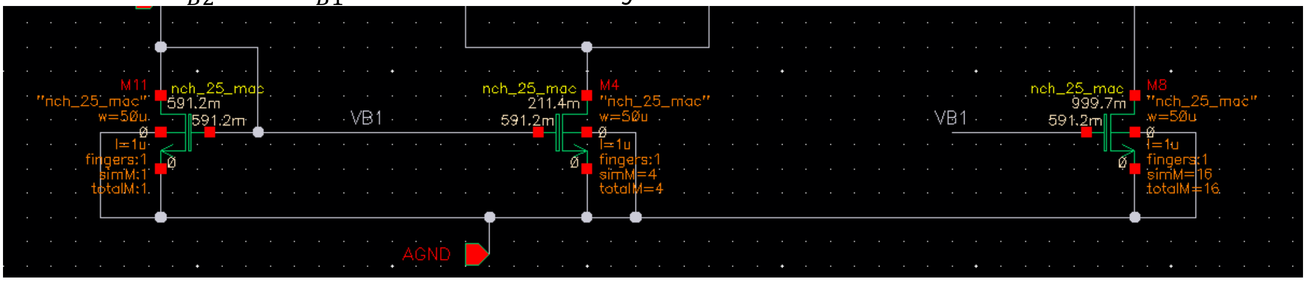

Figure 4 Tail Current Mirror

Second Stage CS Amplifier:

$$\boldsymbol{I_D = I_{B2} = 320\mu A}$$

And Putting $V_{DS} = \frac{V_{DD}}{2} = \mathbf{0.9}\boldsymbol{V}$
And Using ro from the previous device to calculate the gain and get ro for the output device

$$A_v = 34dB \rightarrow 50 \approx \mathrm{gm_{output}}(ro//143K) \rightarrow \boldsymbol{ro = 10K\Omega}$$

And using the same value of VGS for the PMOS Current Mirror Load to avoid latching:

$$\boldsymbol{V_{GS} = 681mV}$$

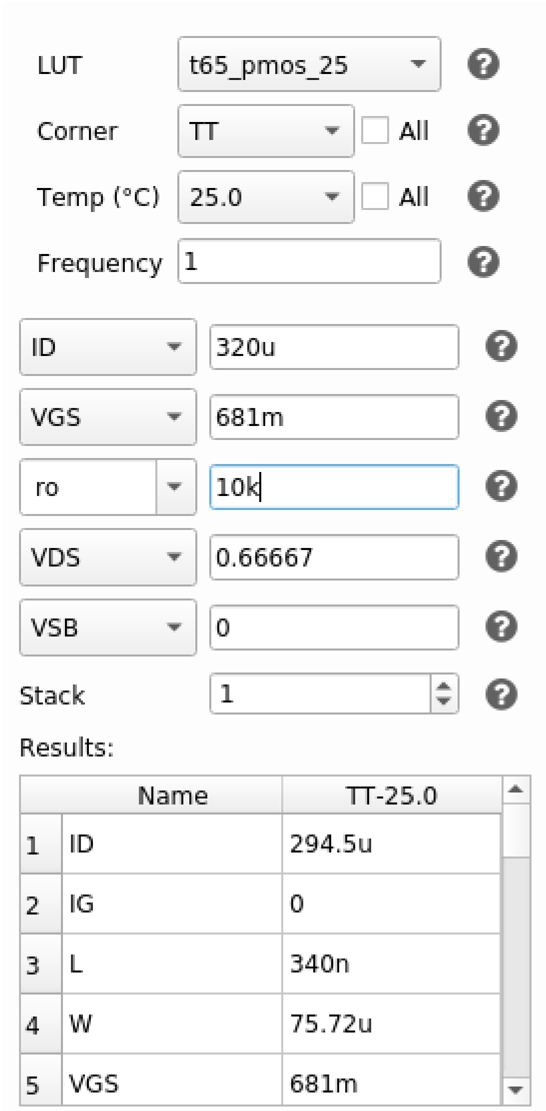

Figure 5 Output PMOS device Sizing from ADT

$$L = 340nm \quad , \quad W = 75.72\mu m$$

Feedforward Zero:

Finally, finding the value of the resistor to cancel the feedforward zero is simple enough:

$$R_C = \frac{1}{gm_{\text{output}}} \approx 300\Omega$$

Initial Design Point:

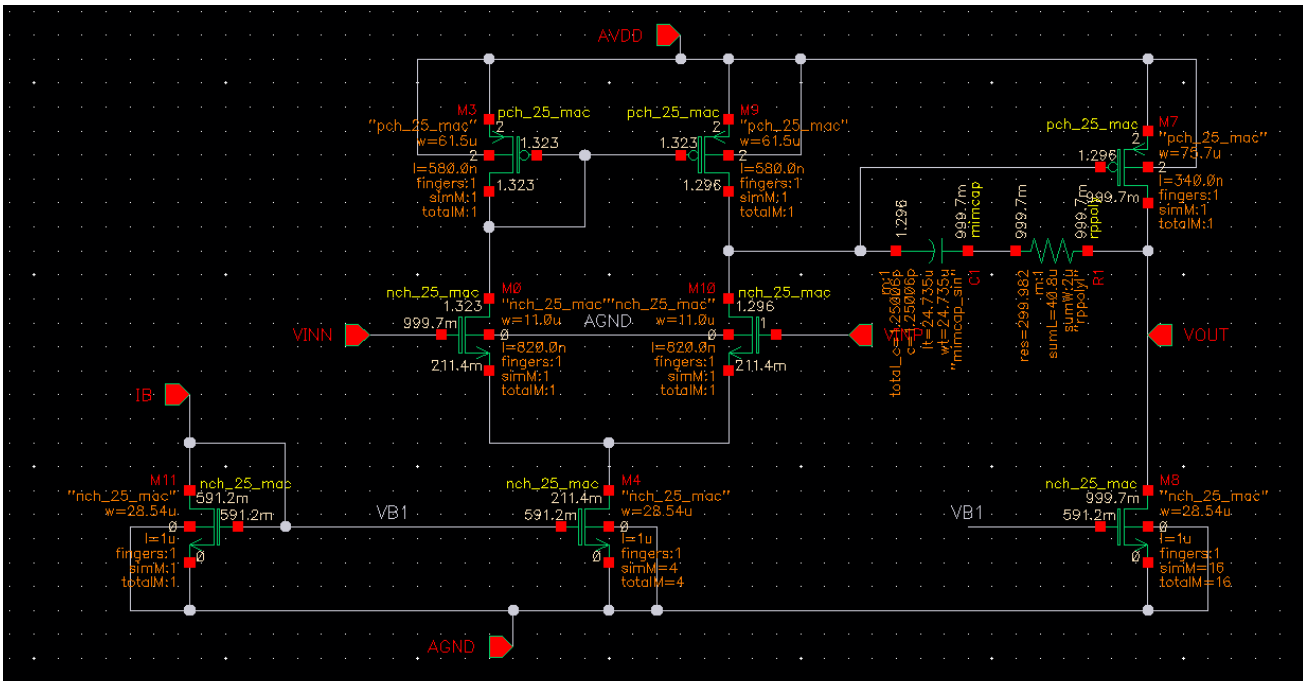

Figure 6 Final Design Schematic with initial sizing of all devices

Simulation and Design Iteration:

The results obtained from the first design point were satisfactory but below the required spec, so I had to do some fine tuning. I increased the widths of the input devices of both stages to get better LG as well as some devices lengths and widths for more accurate results.

I tested the OP point as well as some open loop parameters first in an Open Loop testbench for my design iterations to take a good understanding of what to expect with the circuit before I move to closed loop analysis in the next part

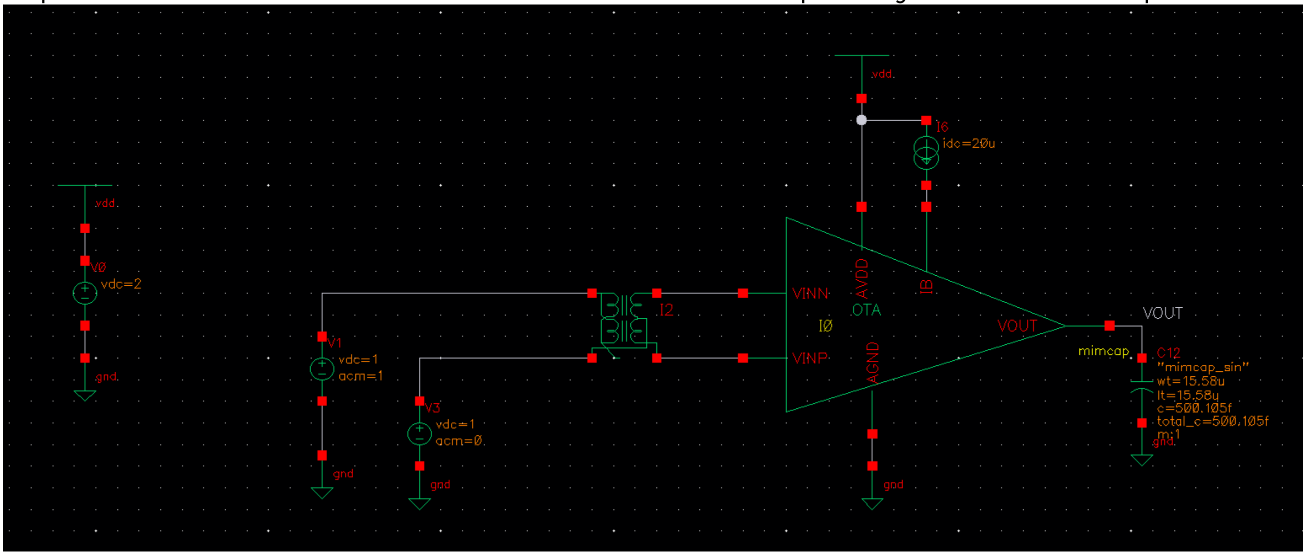

Figure 7 Initial Open Loop Testbench

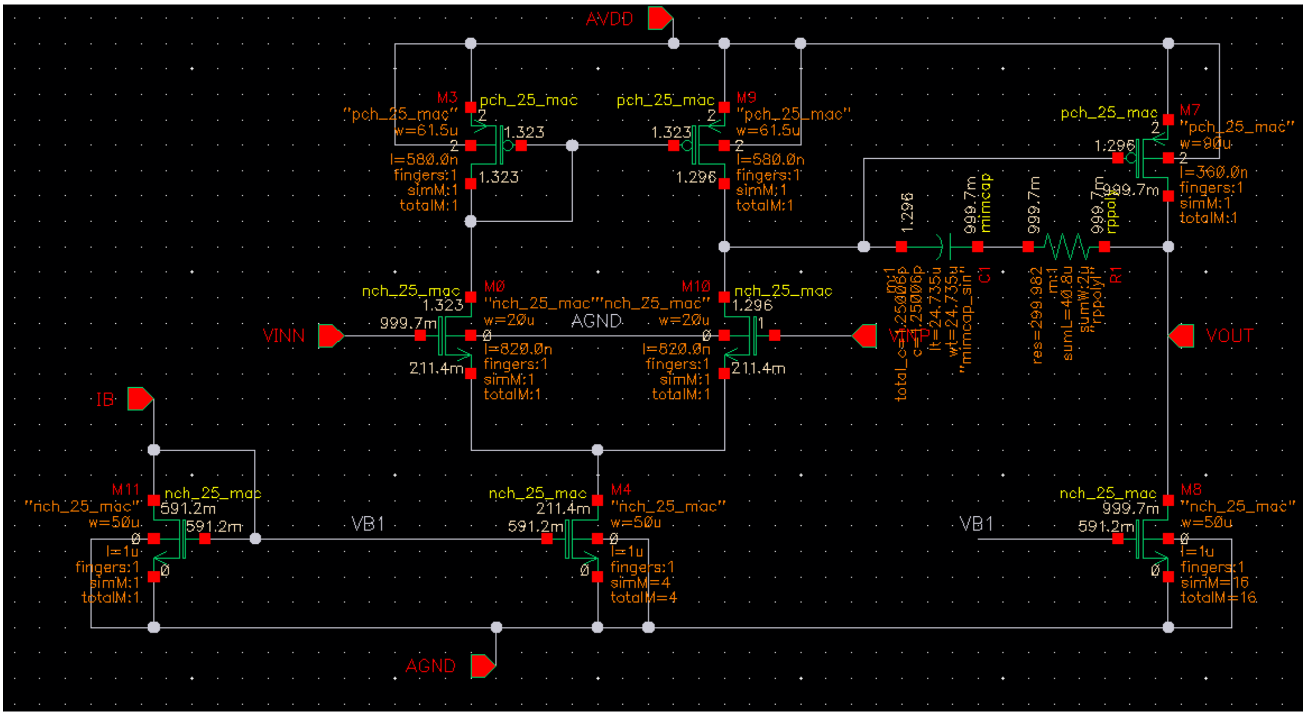

Figure 8 Final Sizing for all devices

Testbench:

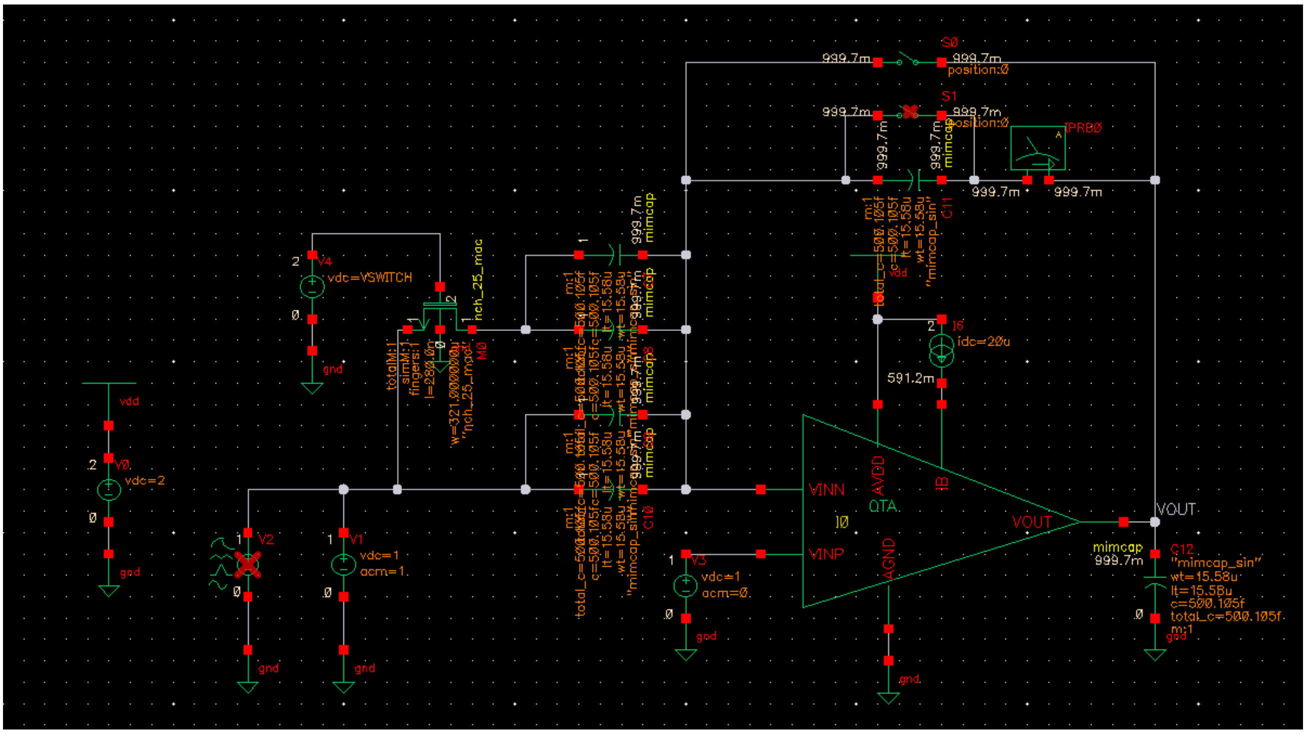

Figure 9 Closed Loop Testbench

As required, I used unit capacitors of 500fF each and configured the testbench in the Inverting configuration.

Switch Sizing:

Last Device required to be sized.
I need its turned-on resistance to be a good margin away from the impedance of the capacitor in the worst-case scenario

$$Ron \ll X_C(@10MHz)$$

$$Assum\ Ron = \frac{X_C}{1000} = \frac{1}{2\pi f C * 1000} \approx 16\Omega$$

Using Minimum L and VDS and VGS = 2V
And using an NMOS device for the switch to be turned on when VGS is equal to VDD (2v: Digital High)

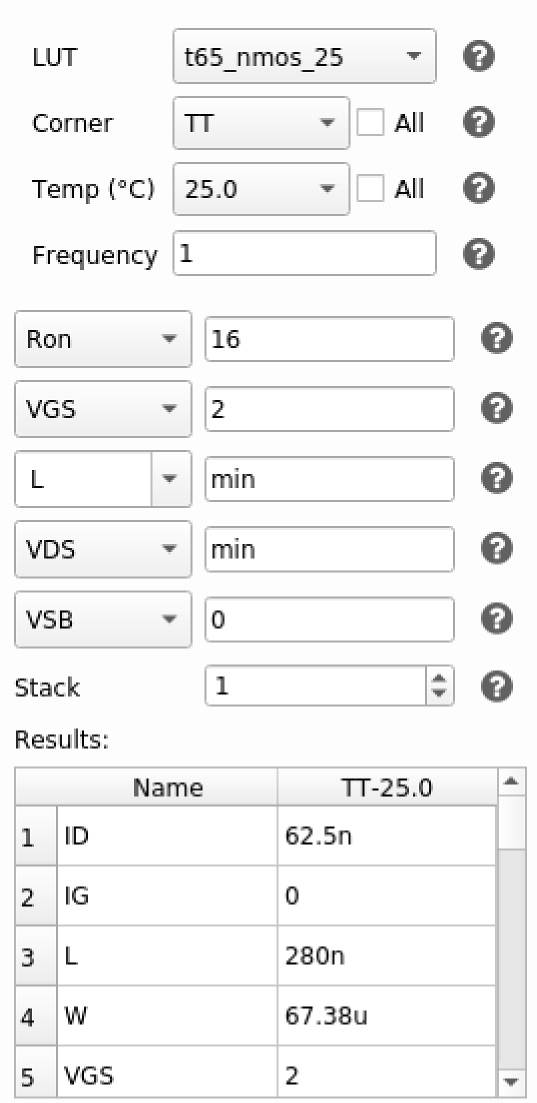

Figure 10 Sizing of Switch from ADT

During Simulation I used even Larger W for even lower RON

Part 1 (Continued): Simulation:

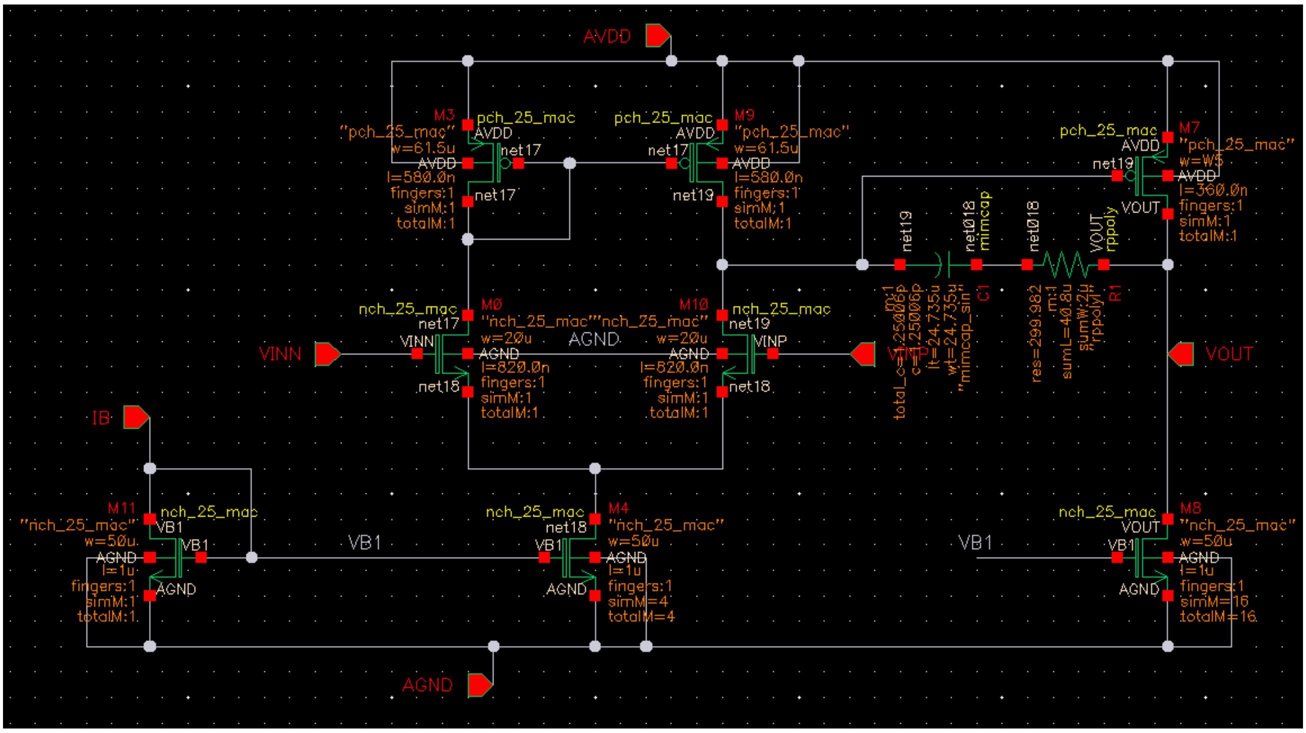

Figure 11 Final Schematic Sizing

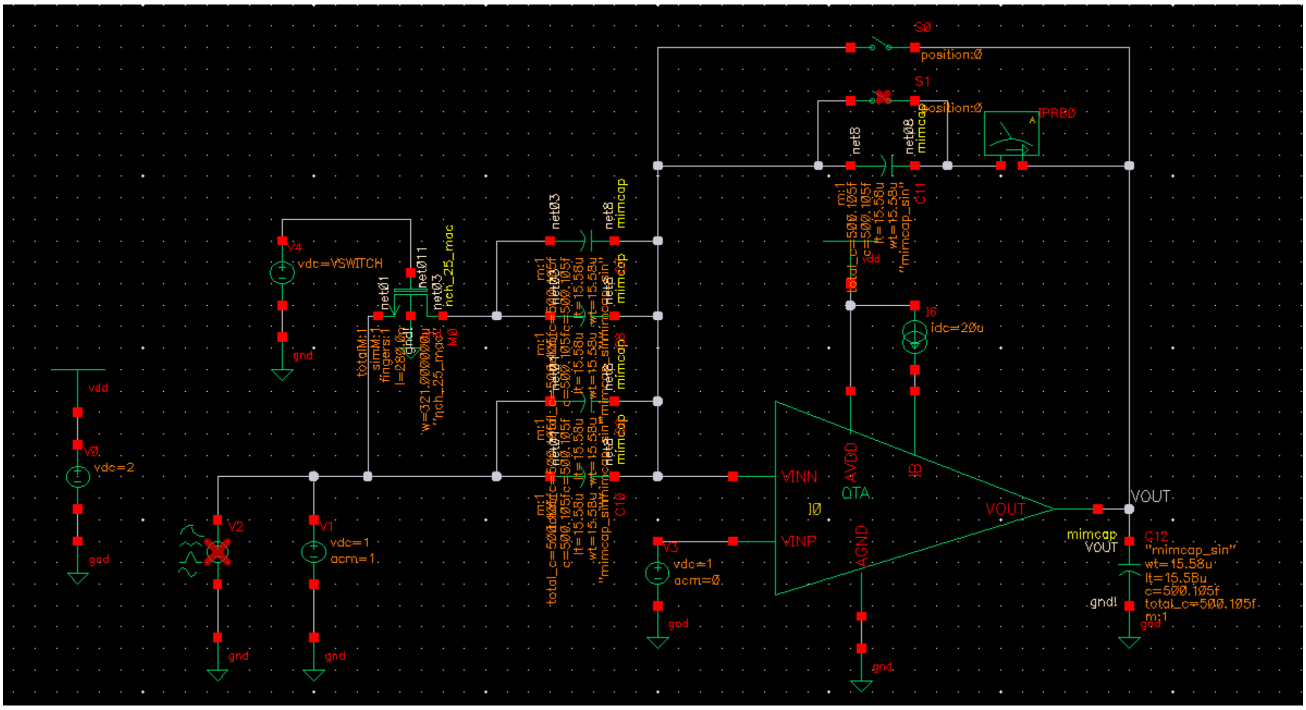

Figure 12 Testbench Schematic

OP Analysis:

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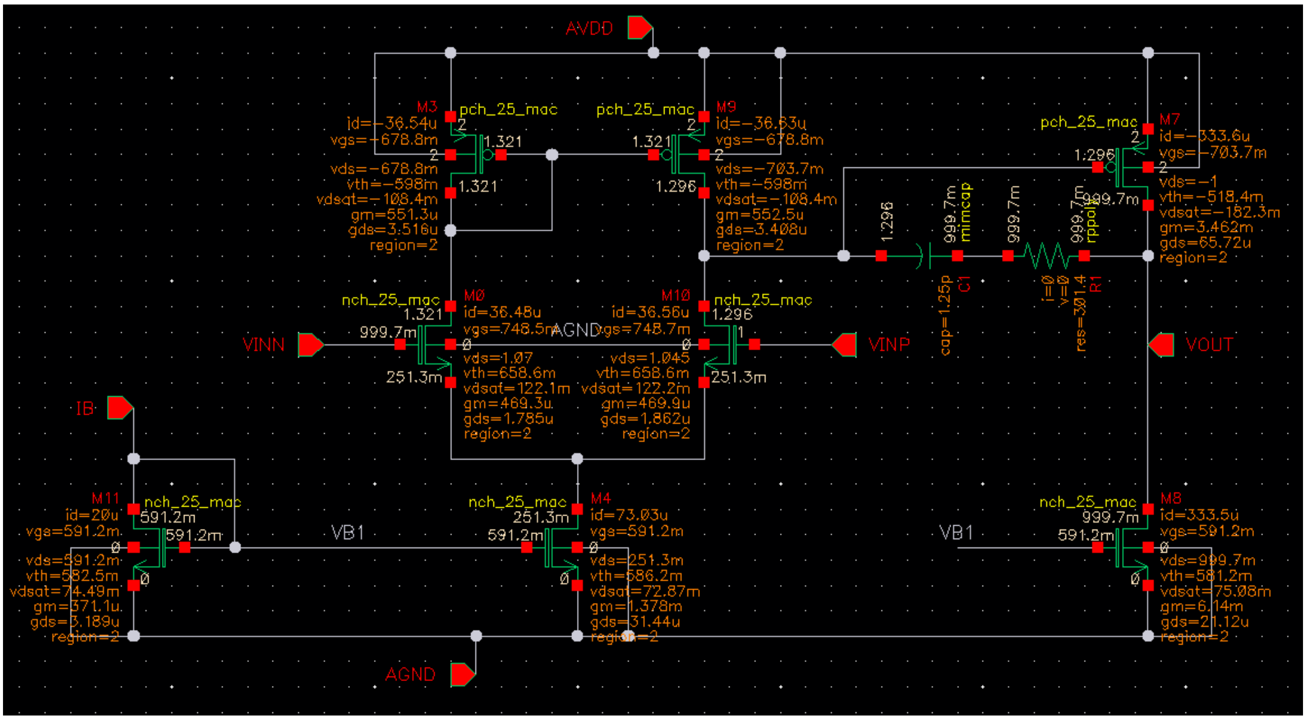

Figure 13 OP Point and DC Voltages Annotated on OTA Schematic

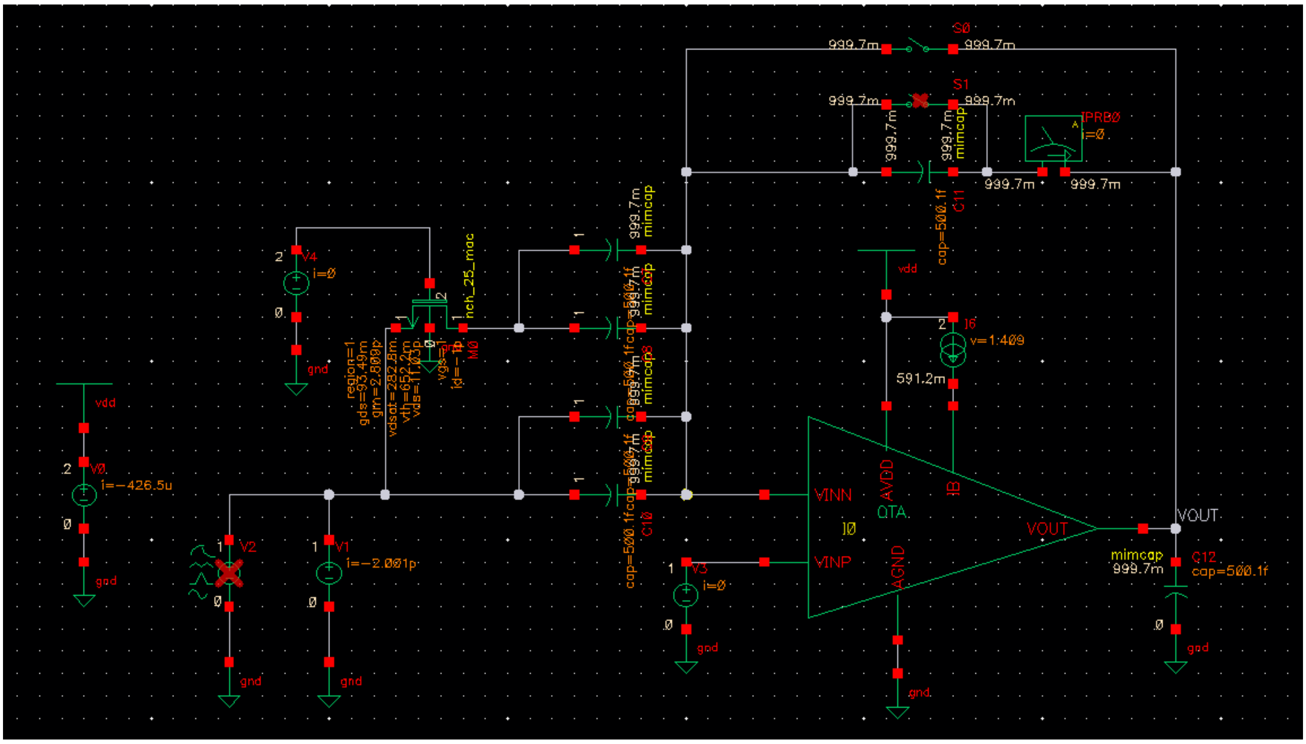

Figure 14 DC Voltages and OP Point annotated on Testbench Schematic

Phase Margin At worst case scenario:

Phase margin is worst in unity gain configuration; I simulated this with a switch shorting the Feedback Capacitor and switch is switched on:

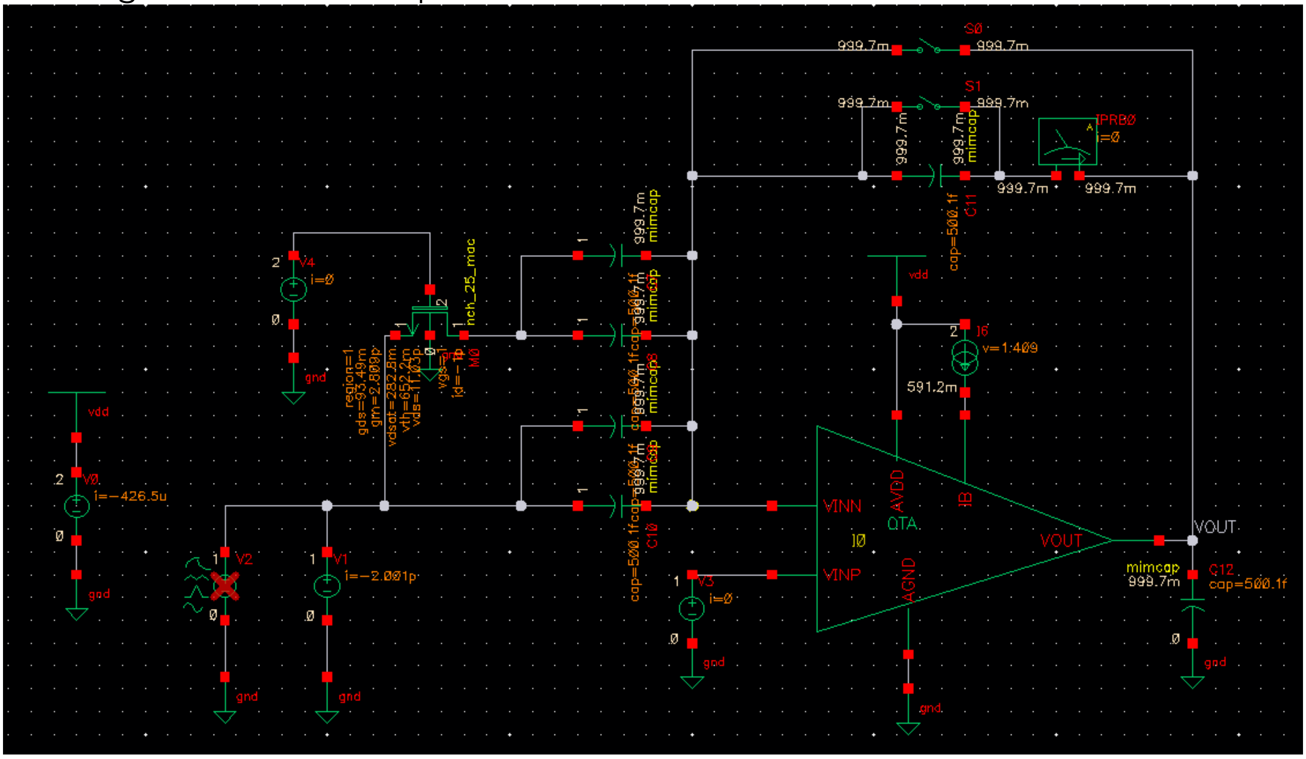

Figure 15 Testbench Configured for Unity Gain Configuration Sim

Running STB Simulation from 1Hz to 10GHz with 10 Points per decade:

Test	Output	Nominal	Spec	Weight	Pass/Fail
DesignChallenge01:Testbench:1	PM	66.69	> 60		pass

Figure 16 Phase Margin from Simulation

$$PM = 66.69^o$$

Phase Margin Meets the Spec!

CL Gain, BW, LG, and PM:

Running both STB and AC analysis from 1Hz to 10GHz
When Switch is closed (D0:1): Worst Case Scenario for LG

Parameters: VSWITCH=0						
1	DesignChallenge01:Testbench:1	Gain				
1	DesignChallenge01:Testbench:1	LG				
1	DesignChallenge01:Testbench:1	BW CL	16.53M	> 10M		pass
1	DesignChallenge01:Testbench:1	ACL Gain	2.075			
1	DesignChallenge01:Testbench:1	LG dB	60.22	> 54		pass
1	DesignChallenge01:Testbench:1	PM	82.95	> 60		pass
Parameters: VSWITCH=2						
2	DesignChallenge01:Testbench:1	Gain				
2	DesignChallenge01:Testbench:1	LG				
2	DesignChallenge01:Testbench:1	BW CL	11.91M	> 10M		pass
2	DesignChallenge01:Testbench:1	ACL Gain	3.994			
2	DesignChallenge01:Testbench:1	LG dB	56.87	> 54		pass
2	DesignChallenge01:Testbench:1	PM	84.42	> 60		pass

Figure 17 Simulation Results

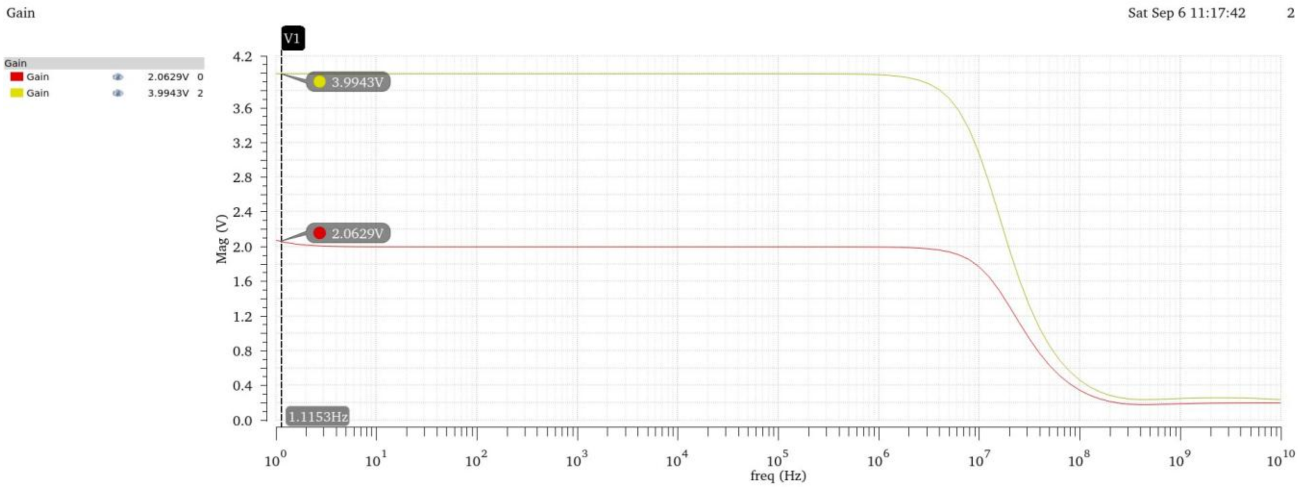

Figure 18 ACL Gain Magnitude

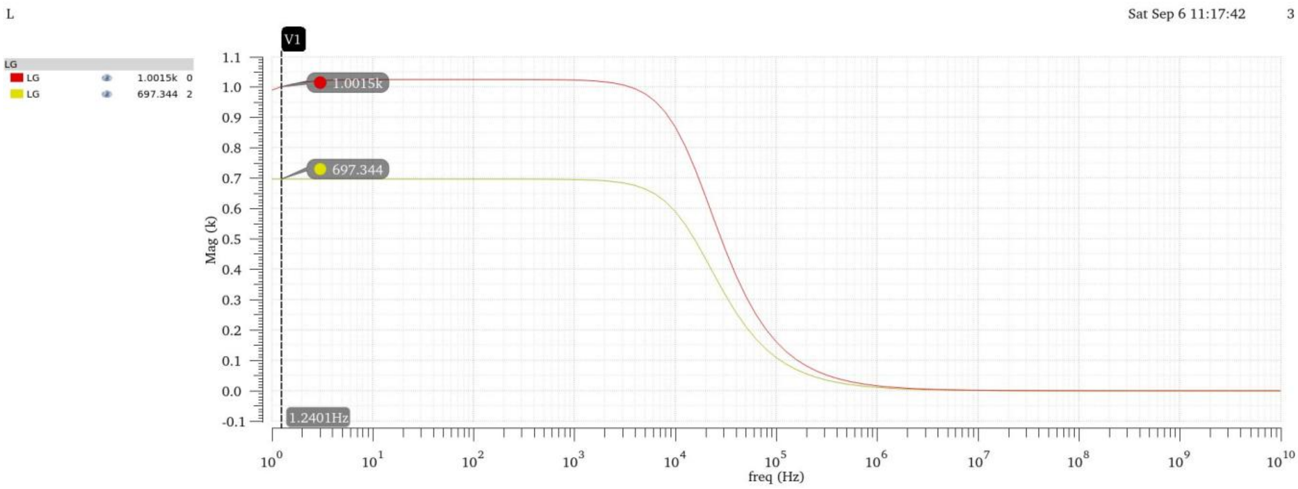

Figure 19 LG in Magnitude

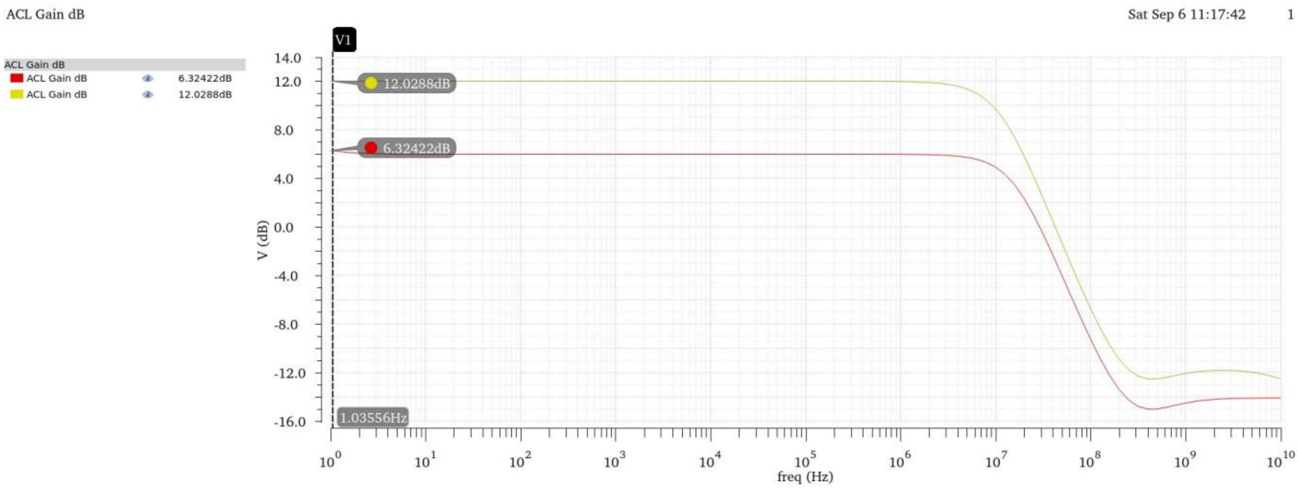

Figure 20 ACL in dB

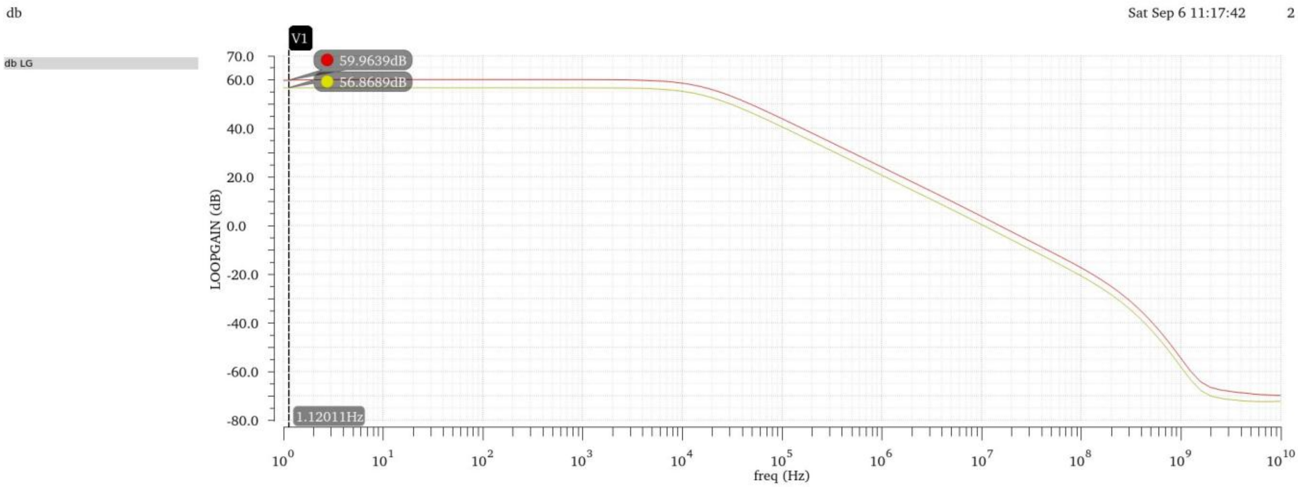

Figure 21 LG in dB

All Specs are Met!

Transient Simulation:

Running Transient simulation using the required source parameters,
Stop Time = 4us and step 10ns

Choosing Peak-To-Peak voltage 0.8V for Gain = 2 Config and 0.4V for Gain = 4 to achieve maximum Swing:

Parameters: VSWITCH=0, VIN=400m				
2	DesignChallenge01:Testbench:1	Transient Voltage Output		
2	DesignChallenge01:Testbench:1	pktopk output	1.596	
2	DesignChallenge01:Testbench:1	Transient Voltage Input		
2	DesignChallenge01:Testbench:1	pktopk Input	800m	
Parameters: VSWITCH=2, VIN=200m				
3	DesignChallenge01:Testbench:1	Transient Voltage Output		
3	DesignChallenge01:Testbench:1	pktopk output	1.591	
3	DesignChallenge01:Testbench:1	Transient Voltage Input		
3	DesignChallenge01:Testbench:1	pktopk Input	400m	

Figure 22 Simulation Results

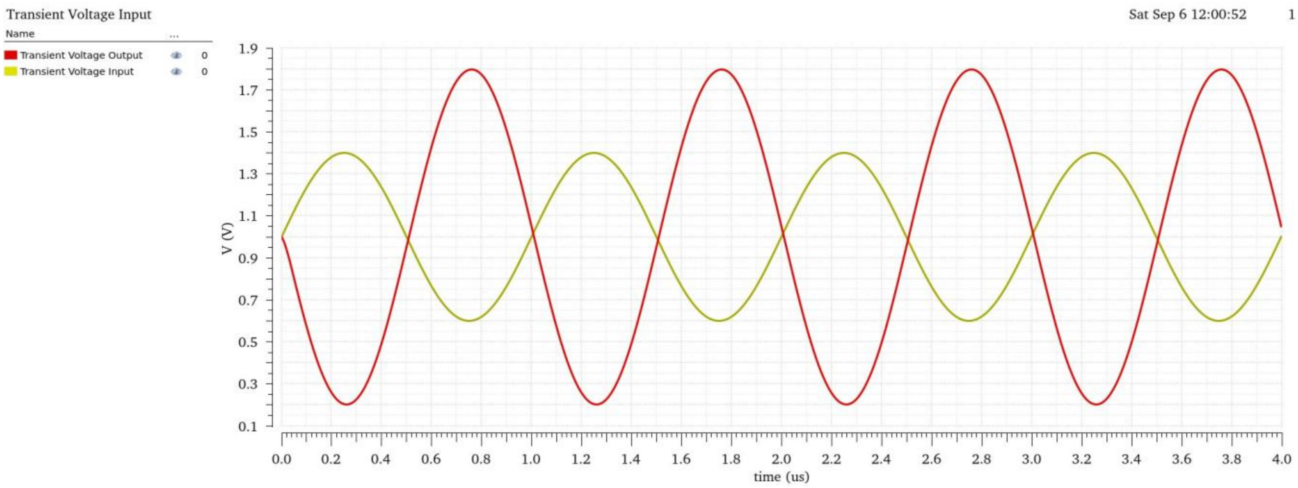

Figure 23 Transient Output and Input Overlaid at Gain = 2 Config

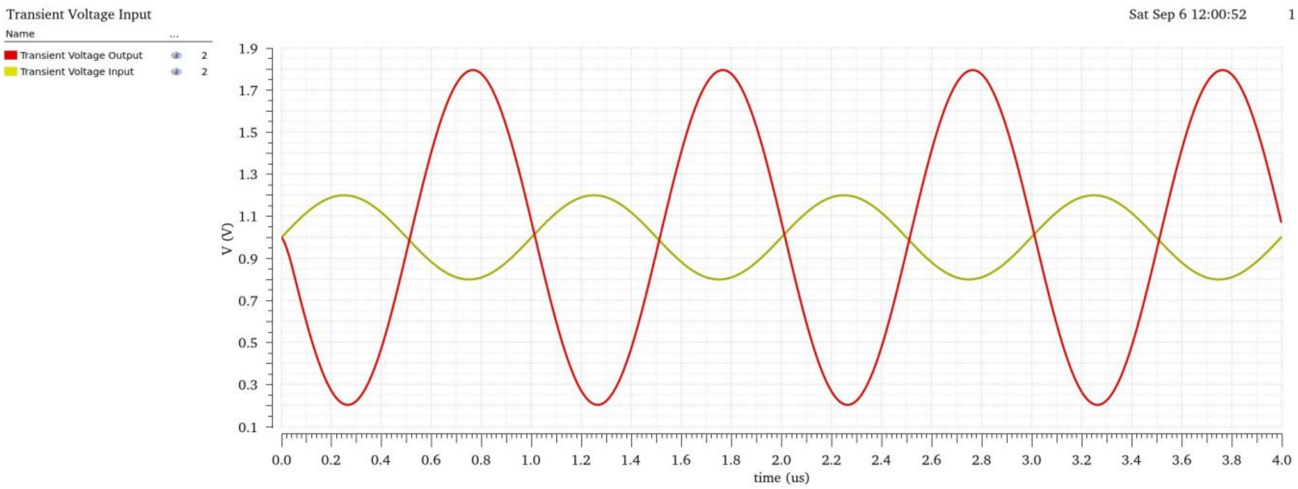

Figure 24 Transient Output and Input Overlaid at Gain = 4 Config

Output swing spec is met!

Part 1: Design Conclusion

Spec	Required	Achieved
Closed loop gain	D0 = 0: 6 dB D0 = 1: 12 dB	D0 = 0: 6.34 dB D0 = 1: 12.028 dB
LG PM (Worst Case)	> 60°	66.69°
DC LG (Worst Case)	> 54 dB	56.87 dB
Closed loop BW	> 10 MHz	11.91 MHz
Output swing	> 1.6 V pk-to-pk	1.591 V pk-to-pk
Power Consumption	Minimize	403uA

All specs appear to be met, Albeit Current Consumption is on the higher side, this is due to the High-Speed requirement of the circuit thus high GBW, Current could be optimized to be a little less than what was achieved but not by a huge margin so I opted to keep it as it is to avoid certain complications in other circuit parameters that have stricter requirements.

Given more time I'd revisit this design and decrease power consumptions by designing a current mirror more accurately and for minimal current consumption as well as attempting to decrease the areas of each device to achieve smaller area instead of aggressively increasing widths to improve certain specs albeit this gives higher margin for the device to behave well across corners.

Part 2: THD and Corners

Total Harmonic Distortion:

Using the previous transient simulation to calculate THD with the following settings:

THD	expr	thd(VT("/VOUT") 1e-06 3e-06 100 1000000)

Figure 25 THD Output setup

SWITCH=0, VIN=400m		
DesignChallenge01:Testbench:1	THD	131.3m
SWITCH=2, VIN=200m		
DesignChallenge01:Testbench:1	THD	189.4m

Figure 26 Simulation Results

THD Appears to be Minimal!

Corner Simulation:

I had already replaced all passives with PDK passives throughout this entire report.

Corners	Nominal	ss	ff
Temperature			
Design Variables			
temperature		100	-40
Click to add			
Parameters			
Click to add			
Model Files			
...s_2d5_lk_v1d0.scs		<section>	<section>
...2d5_lk_v1d0.scs:1		ss_25	ff_25
...2d5_lk_v1d0.scs:2		ss_mim	ff_mim
...2d5_lk_v1d0.scs:3		ss_res	ff_res
...2d5_lk_v1d0.scs:4		<section>	<section>

Figure 27 Corner Setup

Optimally corners should be ran across all device combinations and across all temperatures and VDD Variations, but for this challenge I will run only across the 2 requested corners.

DC LG, ACL and BW across Corners:

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ss	ff
Parameters: VSWITCH=0, VIN=400m										
2	DesignChallenge01:Testbench:1	Gain								
2	DesignChallenge01:Testbench:1	LG								
2	DesignChallenge01:Testbench:1	ACL Gain dB								
2	DesignChallenge01:Testbench:1	db LG								
2	DesignChallenge01:Testbench:1	BW CL	16.53M	> 10M		pass	13.92M	19.88M	13.92M	19.88M
2	DesignChallenge01:Testbench:1	ACL Gain	2.075				2.06	2.096	2.06	2.096
2	DesignChallenge01:Testbench:1	LG dB	60.22	> 54		pass	59.2	61.43	61.43	59.2
2	DesignChallenge01:Testbench:1	PM	82.95	> 60		pass	82.02	83.78	83.78	82.02
Parameters: VSWITCH=2, VIN=200m										
3	DesignChallenge01:Testbench:1	Gain								
3	DesignChallenge01:Testbench:1	LG								
3	DesignChallenge01:Testbench:1	ACL Gain dB								
3	DesignChallenge01:Testbench:1	db LG								
3	DesignChallenge01:Testbench:1	BW CL	11.91M	> 10M		near	9.785M	14.8M	9.785M	14.8M
3	DesignChallenge01:Testbench:1	ACL Gain	3.994				3.994	3.995	3.995	3.994
3	DesignChallenge01:Testbench:1	LG dB	56.87	> 54		pass	55.97	57.98	57.98	55.97
3	DesignChallenge01:Testbench:1	PM	84.42	> 60		pass	83.6	85.17	85.17	83.6

Figure 28 Closed Loop parameters across Corners

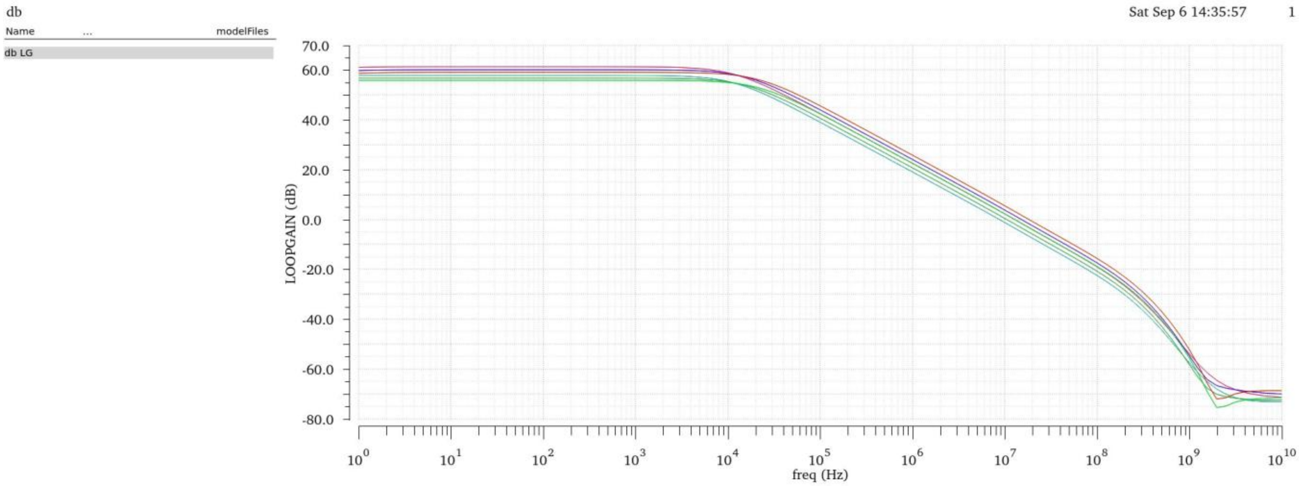

Figure 29 Loop Gain Across Corners and Configurations

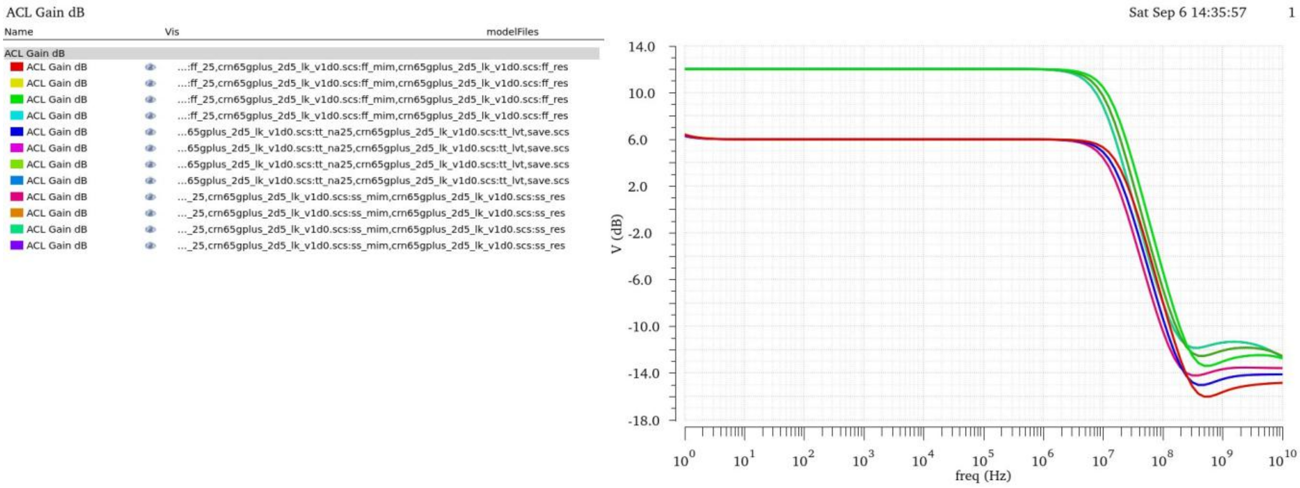

Figure 30 Closed Loop Gain Across Corners and Configurations

All Parameters seem to be satisfied across corner and configurations except for the BW in the Slow Corner but it is still acceptably near.

Phase Margin Across Corners:

I turned on the switch shunting the feedback capacitor again to measure the PM at the worst case across corners and configurations:

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ss	ff
Parameters: VSWITCH=0, VIN=200m										
1	DesignChallenge01:Testbench:1	PM	70.35	> 60		pass	66.94	73.11	73.11	66.94
Parameters: VSWITCH=0, VIN=400m										
2	DesignChallenge01:Testbench:1	PM	70.35	> 60		pass	66.94	73.11	73.11	66.94
Parameters: VSWITCH=2, VIN=200m										
3	DesignChallenge01:Testbench:1	PM	66.69	> 60		pass	63.47	69.3	69.3	63.47
Parameters: VSWITCH=2, VIN=400m										
4	DesignChallenge01:Testbench:1	PM	66.69	> 60		pass	63.47	69.3	69.3	63.47

Figure 31 Phase Margin Across Corners and configurations

Transient Parameters across Corners

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ss	ff
DesignChallenge01:Testbench:1	Transient Voltage Output								
DesignChallenge01:Testbench:1	pktopk output	1.596				1.594	1.596	1.594	1.596
DesignChallenge01:Testbench:1	Transient Voltage Input								
DesignChallenge01:Testbench:1	pktopk Input	800m				800m	800m	800m	800m
DesignChallenge01:Testbench:1	THD	131.3m				113.5m	149.2m	149.2m	113.5m

Figure 32 THD and PkToPk across Corners Gain = 2 Config

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ss	ff
DesignChallenge01:Testbench:1	Transient Voltage Output								
DesignChallenge01:Testbench:1	pktopk output	1.591				1.589	1.593	1.589	1.593
DesignChallenge01:Testbench:1	Transient Voltage Input								
DesignChallenge01:Testbench:1	pktopk Input	400m				400m	400m	400m	400m
DesignChallenge01:Testbench:1	THD	189.4m				161.7m	216.6m	216.6m	161.7m

Figure 33 THD and PkToPk across corners Gain = 4 Config

THD Appears to be minimal across corners except in the SS corner where it is a bit higher

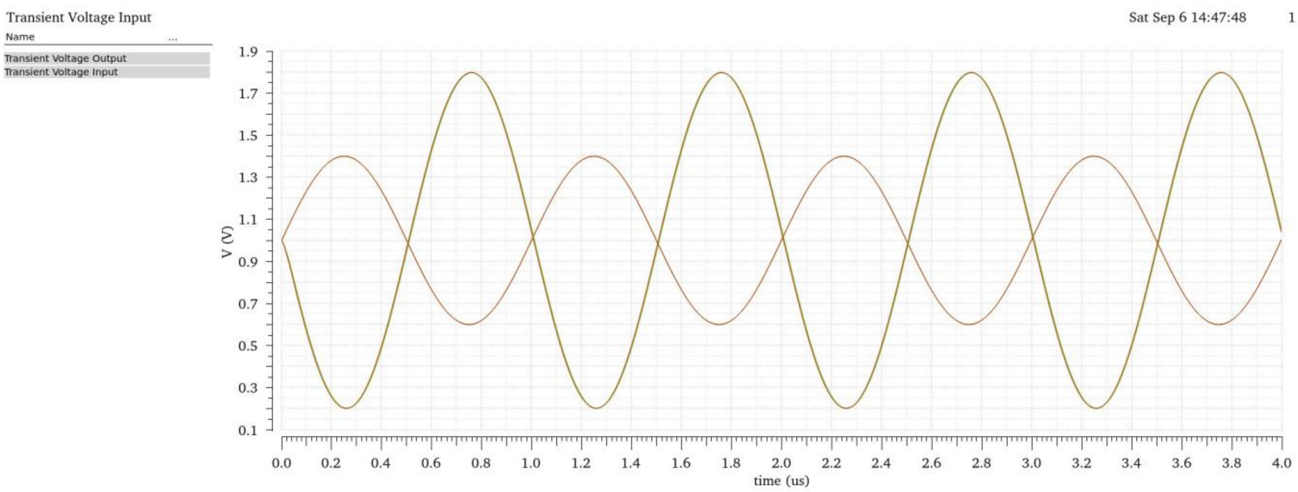

Figure 34 VOUT and VIN overlaid across corners gain = 2

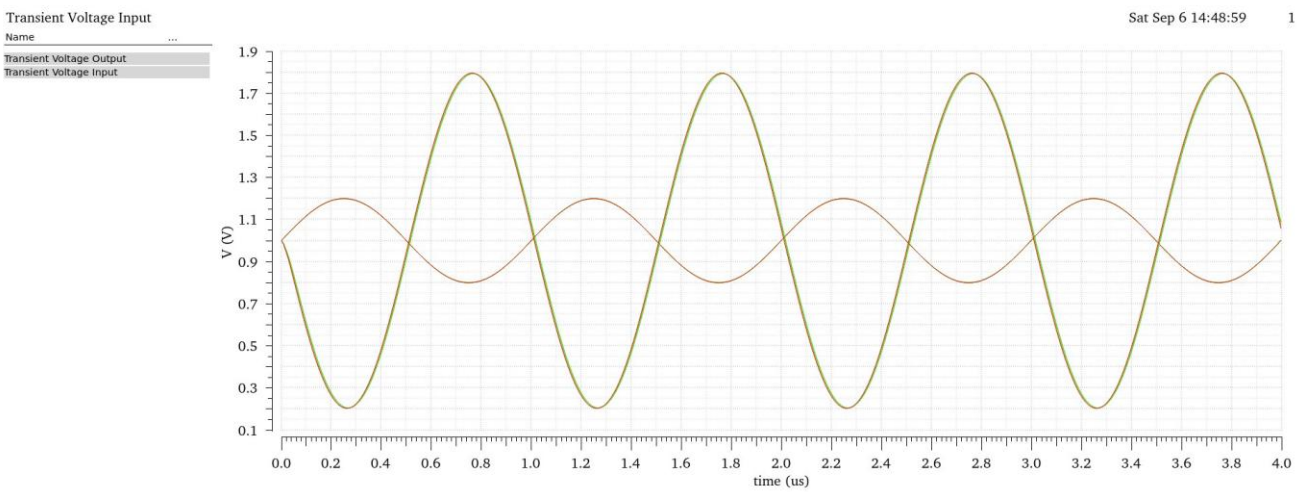

Figure 35 VOUT and VIN overlaid across corners gain = 4

No Visible distortion and all results are overlaying each other perfectly!